Who Pays for the Next Attempt? Contract Design and Escalation Rules for AI-Assisted Software Maintenance

Keywords: AI–human task allocation; escalation; outcome-based pricing; IT service contracts; software maintenance; large language models; double marginalization; optimal stopping

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Abstract

Firms increasingly buy AI-based defect repair as a pay-per-use service: a large language model proposes patches, the firm verifies them, and unresolved defects are escalated to human engineers. We study how the vendor's pricing (per attempt or per verified fix) and the firm's escalation rule interact when defects differ in how easily they are repaired, so that every failed attempt is informative about the next. The firm's optimal rule is a reservation threshold, $N^* = \lceil aH/c - a - b \rceil$ under Beta-distributed difficulty, and the threshold structure holds for any heterogeneity distribution. Per-attempt prices above cost make the firm stop too early (double marginalization). A success fee of $Hk/(k+v)$ restores the efficient depth, but a monopolist sets a higher fee. Deeper sampling requires leaving the firm an *incentive rent of sequential search*, 78–82% of surplus in our calibrations, so the vendor samples too little. Among linear pay-per-use contracts the pure success fee is optimal for the vendor, and a lump-sum fee would remove every distortion. When vendors privately know their quality, only success fees can signal it, and a high-quality vendor separates by lowering its fee, which raises welfare. In a differentiated-Bertrand game, competition removes the distortion only when vendors are close substitutes. On open repair data (RepairBench: 574 bugs, 45 models; a new 40-sample HumanEvalFix experiment) difficulty is heterogeneous in all 84 model–dataset combinations, and a threshold fitted on half the bugs performs within 0.4–1.7% of the best cap in hindsight. Calibrating costs to public wage statistics, task-time annotations, and API price lists shows that the conclusions are strongest when verification is cheap relative to human repair; a field-validation design is provided.

1. Introduction

Software maintenance absorbs most of the lifetime cost of information systems (Lientz and Swanson 1980), and LLM-based repair agents change how that work is organized. A firm can route a defect first to an AI service, accept a patch that passes its tests, and escalate to a person only if the AI fails. Research on AI augmentation stresses that such arrangements succeed or fail on the allocation of tasks and the design of delegation (Raisch and Krakowski 2021; Fügener et al. 2022), but two questions that arise concretely have been left open: how many failed attempts should a firm tolerate before escalating, and how does the vendor's price structure shape that choice?

Both questions turn on a regularity that is easy to overlook. Repair is stochastic, so a defect can be attempted repeatedly, but defects differ enormously in how likely an attempt is to succeed: some are fixed almost every time, others almost never. A run of failures is therefore *bad news about the defect itself*, and the marginal value of another attempt falls as failures accumulate. A firm that plans with an average success rate would never stop; a firm that fixes the number of attempts by convention ignores both its own cost of human repair and the vendor's price.

Pricing matters because it sets the cost of an additional attempt as the firm sees it. Under *per-attempt* pricing (per call or per token), the firm bears the price of every attempt, including the many that fail on hard defects. Under *outcome-based* pricing, it pays only for a verified fix and the vendor bears the compute cost of failures.

Contributions. (1) We model AI–human escalation as a sequential search problem with an outside option inside a vendor–buyer game, linking the AI-delegation and IT-contracting literatures. (2) We show that per-attempt pricing above cost causes double marginalization, that a success fee of $Hk/(k+v)$ restores efficiency, and that a monopolist nevertheless sets a higher fee and samples too little (Propositions 2–4). (3) We identify the underlying force, an *incentive rent of sequential search*, and prove, for any heterogeneity distribution, that the vendor's optimal pay-per-use contract is a pure success fee and that a

lump-sum fee removes all distortions (Propositions 5–6). (4) We show that private vendor quality reverses profit order across contract forms so that only success fees can signal quality (Proposition 7), and that competition removes the distortion only among close substitutes (Proposition 8). (5) We test the model's premises on open repair data and report numerical equilibria calibrated to them.

Article type and audience. This is a theory article in the ISR sense: it applies and extends search theory and vertical-contracting theory to a new IS phenomenon (pay-per-use AI services), derives propositions, and illustrates the model with open repair data. The intended audience is academics working on the economics of information systems, IT sourcing and governance, and human–AI work design, and practitioners who procure or price AI-assisted software services (chief information officers, engineering and procurement managers, and AI vendors).

Roadmap. Section 3 sets up the model and a running example. Section 4 derives the firm's escalation rule, the monopoly contracts, the incentive rent, and the effects of private vendor quality and competition. Section 5 reports numerical equilibria and a sensitivity analysis with costs calibrated from public sources. Section 6 tests the model's premises on open repair data. Sections 7–9 discuss extensions, implications, and limits.

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2. Related Literature and Theoretical Positioning

IT contracting and outsourcing. Information systems research has studied how clients and vendors contract for software and IT services. Whang (1992) models contracting for software development as a game in which the user knows the value of the system better than the developer and the developer learns its costs as work proceeds. Gopal et al. (2003) analyze the choice between fixed-price and time-and-materials contracts in offshore software projects: requirement uncertainty, team size, and resource shortage explain the choice, and the choice affects vendor profit, with the main risk borne by the vendor under fixed price and by the client under time and materials. Chen and Bharadwaj (2009) study contract structures in IT outsourcing through transaction cost, agency, and relational exchange theories, and

Susarla et al. (2010) study contract extensiveness, duration, and extendibility as responses to holdup under incomplete contracts. Bajari and Tadelis (2001) give the procurement-economics counterpart: because of costly ex post renegotiation, cost-plus contracts are preferred to fixed-price contracts when projects are more complex. These papers compare contracts that pay for inputs (time and materials, cost-plus) with contracts that pay for a deliverable (fixed price). Per-attempt and per-result pricing of AI services has the same input/output structure, with two differences that change the analysis: the input is a *stochastic attempt* whose failure carries information about the next, and the buyer, not the vendor, decides sequentially how many attempts to purchase. The relevant friction is therefore the buyer's continuation decision, and the quantity we study is an *incentive rent* in that decision, not the risk allocation or renegotiation costs emphasized in the contracting papers.

Agency and incentives. Agency theory frames contract choice in terms of information, outcome uncertainty, incentives, and risk (Eisenhardt 1989). Holmström and Milgrom (1991) show that when pay both allocates risk and rewards work, and the agent has several tasks, incentives on measurable outcomes can distort effort away from unmeasured dimensions. Our success fee pays for an outcome the buyer can verify with tests; the virtue of paying for results in the model carries this multitask caveat, which we discuss in Section 8.

Human–AI collaboration. IS and management research studies how people work with and delegate to AI. Raisch and Krakowski (2021) argue that automation and augmentation are interdependent and not alternatives. Fügener et al. (2021) show that working with AI can raise individual accuracy while reducing the diversity of human knowledge available to a group. Fügener et al. (2022) find that human–AI teams outperform the AI alone when the AI delegates work to humans but not when humans delegate work to the AI, and that humans delegate poorly. Ibrahim et al. (2021) show how to elicit human judgment so that it improves algorithmic prediction. Experimental and field evidence reports productivity effects of AI assistants for programming and writing (Peng et al. 2023; Brynjolfsson et al. 2025), and Yeverechyahu et al. (2024) study the effect of an LLM coding assistant on open-source activity. This work concerns individual and team performance with AI. We study its organizational counterpart in

repair: the *escalation boundary* at which a firm hands a defect from AI to a person, and how the AI vendor's price structure moves that boundary. Since humans delegate imperfectly (Fügener et al. 2022), a rule based on observable quantities, such as a threshold computed from past success rates and costs, is a natural design.

Pricing of information goods and search. Pricing structure matters for profit in information goods: Sundararajan (2004) shows that adding a fixed fee to a nonlinear usage-based scheme can raise profits when transaction costs exist, and Bakos (1997) shows that information systems lower buyers' search costs and thereby sellers' market power. Our lump-sum result (Proposition 6) and the role of differentiation in competition (Proposition 8) are counterparts in the AI-services setting. The sequential structure draws on search theory (Stigler 1961; McCall 1970; Weitzman 1979); the rent we identify relates to the information rents of optimal mechanism design (Myerson 1981) and the distortion to double marginalization in vertical relations (Spengler 1950; Tirole 1988).

Software maintenance and automated repair. Maintenance absorbs a large share of lifecycle effort (Lientz and Swanson 1980), and its cost depends on system structure and volatility (Kemerer and Slaughter 1999; Banker and Slaughter 2000). LLM-based repair is evaluated by tests and reported as pass@k (Chen et al. 2021; Le Goues et al. 2012; Xia et al. 2023; Jimenez et al. 2024; Silva and Monperrus 2025), and repeated sampling raises coverage with diminishing returns (Brown et al. 2024). These papers measure capability, not the economic choice of how many attempts to buy; for the contribution of AI to economic activity more broadly see Agrawal et al. (2018).

Theoretical positioning. We apply and extend existing theory: contract and agency theory (what to pay for), search theory (when to stop), and vertical contracting (how a seller's price distorts a buyer's quantity). The new theoretical content is (i) the incentive rent of sequential search, which holds for any distribution of repair difficulty and explains the optimal pay-per-use contract and the under-supply of sampling; (ii) the reversal of profit order across contract forms under private vendor quality, which makes outcome-based pricing the only credible signal; and (iii) the role of differentiation, more than the number of vendors, in competition. The setting is specific to information systems for three reasons: success can be

verified cheaply by tests, attempts are metered by software at near-zero marginal cost, and a vendor's quality is a private property of a model that buyers cannot observe directly.

Testable implications for IS research. The model yields predictions on contract and usage data. (1) Outcome-based pricing should be more common where vendors are few or differentiated, and per-attempt pricing near cost where vendors are close substitutes (Proposition 8). (2) Buyers for whom human repair is expensive relative to per-attempt verification should tolerate deeper sampling, and observed escalation thresholds should fall with the price per attempt (Proposition 1). (3) Vendors with market power should rely on result-based fees and not on per-attempt fees when repair difficulty is heterogeneous (Proposition 5). (4) High-quality vendors should be more likely to offer outcome-based terms (Proposition 7). (5) Where prepayment or access fees are feasible, observed sampling should be closer to the efficient depth (Proposition 6).

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3. Model

Players and timing. A firm faces a stream of defects; a monopolist vendor offers an AI repair service. (1) The vendor announces a contract. (2) The firm accepts, or rejects and has every defect repaired by a human at cost H. (3) For each defect the firm requests attempts sequentially, verifies each patch at its own cost, and stops on an accepted patch or escalates to a human.

Defects. Each defect has a latent per-attempt success probability $p \sim F$, unobserved by both parties; given p, attempts are independent and their outcomes (pass or fail the firm's tests) are observed by both. Let T be the index of the first success, S(j) = P(T > j) = E[(1−p)^j], and q_n = P(attempt n+1 succeeds | n failures) = E[p(1−p)^n]/E[(1−p)^n]. For F = Beta(a, b), S(j) = B(a, b+j)/B(a, b) and q_n = a/(a+b+n). Let E(N) = Σ_{j<N} S(j) be the expected number of attempts under a cap N and P(N) = 1 − S(N) the probability of repair within N attempts.

Costs. An attempt costs the vendor k and the firm v (tests, triage, review); verification is accurate.

Contracts. (A) *Per-attempt:* the firm pays $w \geq k$ per attempt. (B) *Success fee:* the firm pays $f \leq H$ per verified fix and nothing for failures. The joint benchmark treats the cost per attempt as $k + v$. Total surplus relative to human-only repair for a cap N is

$$W(N) = H \cdot P(N) - (k + v) \cdot E(N),$$

independent of the transfer. N^FB maximizes W. We call (A) and (B) *pay-per-use* contracts, since payments depend only on realized attempts and outcomes.

3.1 A running example

To fix ideas we use one configuration throughout; it is also the base case of the tables below. Repair difficulty has a = 0.12 and b = 0.39 (the median fit to RepairBench in Section 6), a human repair costs H = $100, an attempt costs the firm v = $1 (tests and triage) and the vendor k = $0.20 (compute). Under this distribution the average first attempt succeeds with probability 23.5%, but the distribution is U-shaped: 47% of defects have a per-attempt success probability below 1%, while 22% have one above 50%. Failure is therefore informative. After ten failures the probability that the next attempt succeeds has fallen from 23.5% to 1.1%, because ten failures are strong evidence that the defect belongs to the nearly unsolvable group. In dollars, the first attempt is worth $q_0 H$ = $23.5 to the firm, the second $7.9, the third $4.8, and the tenth $1.26, against a social cost per attempt of k + v = $1.20; the eleventh attempt is worth $1.14. The efficient depth is therefore N^FB = 10 (Figure 1a).

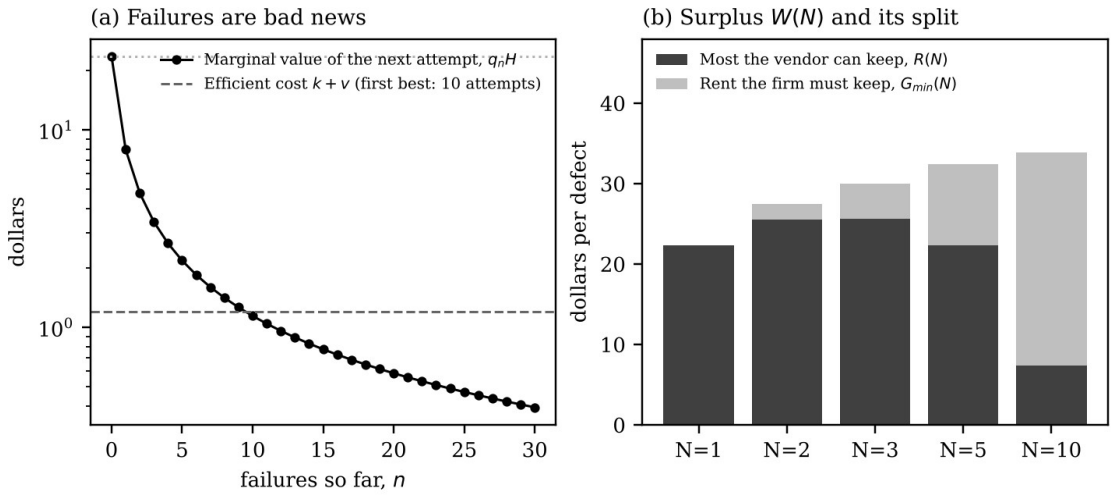


Figure 1. *The running example (a = 0.12, b = 0.39, H = 100, v = 1, k = 0.2). (a) The marginal value of the next attempt, q_n H, falls steeply with the number of failures n and crosses the efficient cost k + v after ten attempts. (b) For each depth N, the bar shows the total surplus W(N) split into the most a vendor can keep, R(N), and the incentive rent the firm must keep, G_min(N) (Section 4.3).*

Table 1. Running example: surplus and its split by attempt budget N (dollars per defect; P(N) is the probability that the AI repairs the defect within N attempts, E(N) the expected number of attempts, W(N) the surplus relative to human-only repair, G_min(N) the rent the firm must keep to be willing to go N attempts deep, and R(N) = W(N) − G_min(N) the most the vendor can keep; Section 4.3).

N	P(N)	E(N)	W(N)	G_min(N)	R(N)
1	0.235	1.00	22.33	0.00	22.33
2	0.296	1.76	27.49	1.96	25.53
3	0.330	2.47	30.01	4.43	25.58
5	0.370	3.79	32.44	10.11	22.33
10	0.420	6.82	33.86	26.50	7.36

Deeper sampling raises the surplus from $22.3 to $33.9 per defect, but the share that a vendor can keep falls from all of it at N = 1 to 22% at N = 10. The vendor's maximum, $25.6, is reached at N = 3. These numbers explain the equilibria in Sections 4 and 5.

4. Results

4.1 The firm's escalation rule

Proposition 1 (Reservation rule). For every non-degenerate F, q_n is strictly decreasing in n. Under contract A the firm makes attempt n+1 iff q_n·H > w + v; under B iff q_n·(H − f) > v. For F = Beta(a, b) the budgets are N_A(w) = max{0, ⌈aH/(w+v) − a − b⌉}, N_B(f) = max{0, ⌈a(H−f)/v − a − b⌉}, and N^FB = max{0, ⌈aH/(k+v) − a − b⌉}.

Proof. Let E_n be expectation under the tilted law ∝ (1−p)^n dF, so q_n = E_n[p] and q_{n+1} = (E_n p − E_n p²)/(1 − E_n p) ≤ E_n p = q_n by Jensen, strictly unless p is degenerate under that law. Since q_n is decreasing, once continuing stops being preferred it never is again, so the one-step look-ahead rule is optimal (Chow, Robbins, and Siegmund 1971). In state n the comparison is stopping (cost H) versus one more attempt and then stopping: (w+v) + (1−q_n)H under A, or v + q_n f + (1−q_n)H under B. ∎

The threshold rises with the cost of human repair, falls with the price of an attempt, and falls with difficulty (low *a*, high *b*). A homogeneous model (constant p) would prescribe never stopping or never starting; heterogeneity is what makes the rule interior.

Intuition and example. The rule says to keep going while one more attempt is worth more than it costs. What makes it informative is that the worth of an attempt falls as failures accumulate. In the running example a firm that used the average success rate would value every attempt at $23.5 and never stop; the true value of the third attempt is $4.8 and of the tenth $1.26. At an effective cost of $1.20 per attempt the firm stops after ten, and at $6 per attempt (a price of $5 plus its own $1) after ⌈12/6 − 0.51⌉ = 2.

4.2 Pay-per-use contracts under monopoly

Proposition 2 (Double marginalization). Under A with w > k, N_A(w) ≤ N^FB, and surplus is below its first-best level whenever w lowers the budget.

Proof. N_A decreases in w, so N_A(w) ≤ N_A(k) = N^FB. W(N+1) − W(N) = S(N)[q_N H − (k+v)] is positive iff N < N^FB, so W increases up to N^FB. ∎

Proposition 3 (Efficient success fee). Under B with f^e = Hk/(k+v), N_B(f^e) = N^FB; the vendor earns k/(k+v) of the first-best surplus.

Proof. (H − f^e)/v = H/(k+v), so the thresholds coincide. Vendor profit is f^e P − kE = (k/(k+v))[HP − (k+v)E]. ∎

Example. In the running example, a per-attempt price of $5 makes the firm stop after two attempts, and the surplus falls from $33.9 to $27.5 per defect (Table 1). A success fee of f^e = 100 × 0.2/1.2 = $16.67 leaves the firm $83.3 per success; since $q_9 \times 83.3 = 1.05$ exceeds v = 1 and $q_{10} \times 83.3 = 0.95$ does not, the firm again stops after ten attempts, and the vendor keeps one-sixth of the $33.9, or $5.6.

Proposition 4 (The monopoly fee exceeds the efficient fee). Let N^FB ≥ 1 and Π(f) = f·P(N_B(f)) − k·E(N_B(f)). Because N_B is an integer step function of f: (a) Π is strictly increasing within each step, and for every f′ ≤ f^e there is f > f^e with Π(f) > Π(f′); (b) the budget at the vendor's optimum satisfies n̂ ≤ N^FB, strictly unless the best fee lies in the same step as f^e, in which case the vendor raises the fee without changing the firm's depth.

Proof sketch (full proof in Appendix E). The firm's budget is n exactly for f ∈ [φ_n, φ_{n−1}), φ_n = H − v/q_n. Let L(n) be the profit at the lower end and R(n) the supremum of profit within step n. Then R(n) − L(n−1) = S(n−1)[q_{n−1}H − (k+v)], which is negative iff attempt n is inefficient. For n > N^FB, R(n) ≤ L(n−1) < R(n−1), so R is strictly decreasing beyond N^FB. ∎

Numerically the vendor's best step is strictly below N^FB in 289 of 293 random draws; in the other four it raises the fee within the efficient step.

4.3 The incentive rent of sequential search

Proposition 5 (Incentive rent and optimal linear contract). For N ≥ 1 define the incentive rent G_min(N) = v·[P(N)/q_{N−1} − E(N)] = v·Σ_{j<N} S(j)·(q_j/q_{N−1} − 1) ≥ 0. (a) G_min(1) = 0, G_min is nondecreasing in N, and G_min ≡ 0 iff difficulty is homogeneous. (b) For every contract with per-attempt fee w ≥ 0 and success fee f ≤ H under which the firm chooses depth N ≥ 1, the vendor's profit satisfies Π(w, f) ≤ R(N) − (w/v)·G_min(N), where R(N) = W(N) − G_min(N) = Σ_{j<N} S(j)[q_j

φ_{N−1} − k], φ_{N−1} = H − v/q_{N−1}. (c) R(N) is approached only by the pure success fee (w = 0, f ↑ φ_{N−1}); a positive per-attempt component strictly lowers profit whenever N ≥ 2 and difficulty is heterogeneous. (d) The vendor's optimal depth maximizes R(N), and R(N+1) − R(N) = S(N)[q_N H − (k+v)] − v·P(N)·(1/q_N − 1/q_{N−1}); hence n̂ ≤ N^FB. For the Beta family the last factor is 1/a.

Proof sketch (full proof in Appendix E). The firm chooses depth N only if q_{N−1}(H−f) > w + v, i.e. f < φ_{N−1} − w/q_{N−1}. Substituting into Π = (w−k)E + fP gives Π < R(N) − w[P/q_{N−1} − E]. Participation holds at the corner (0, φ_{N−1}) because P ≥ q_{N−1}E. ∎

Interpretation. To make the firm willing to try the N-th attempt, whose success probability is low, the vendor must price the marginal success at φ_{N−1} = H − v/q_{N−1}. Every earlier attempt succeeded more often than the marginal one, so the firm keeps an infra-marginal gain, G_min(N). It is zero when difficulty is homogeneous and grows with dispersion. The vendor's profit is surplus net of this rent, so it samples too little, and the pure success fee delivers any depth most cheaply because a per-attempt fee collects revenue from attempts where the firm's marginal gain is smallest.

Example. To make the firm go three attempts deep, the vendor can charge at most $\varphi_2 = H - v/q_2 = 100 - 1/0.0478 = \$79.1$ per verified fix, which leaves the firm \$20.9 on each success, exactly its break-even cost v/q_2 of buying a success with the third attempt. The first two attempts succeed more often, so the firm has gained on them, and its total gain is G_min(3) = \$4.43. To induce ten attempts the vendor must cut the fee to $\varphi_9 = \$20.8$, and the rent grows to \$26.5 of the \$33.9 surplus (Table 1).

A backlog of 1,000 defects (illustrative). Human-only repair costs \$100,000. A monopolist charging per attempt sets $w = q_0 H - v = \$22.5$, so the firm tries once per defect and the AI repairs 235; the vendor earns \$22.3 thousand and the firm saves almost nothing, so the AI looks cheap per call yet adds little for its buyer. A monopolist charging per verified fix sets \$79.1; the firm tries up to three times, the AI repairs 330 defects, the vendor earns \$25.6 thousand, and the firm saves \$4.4 thousand. The efficient policy samples up to ten times and repairs 420 defects, with a joint surplus of \$33.9 thousand, of which the per-attempt price leaves \$22.3 thousand and the success fee \$30.0 thousand.

Proposition 6 (Lump-sum neutralization). If the vendor may also charge a fee T_0 independent of attempts and outcomes, then (w, f, T_0) = (k, 0, W(N^FB)) implements N^FB and gives the vendor the entire surplus; for any (w, f), T_0 can recapture G_min without changing the firm's budget.

Proof. T_0 is a constant in the firm's cost and does not enter any one-step comparison of Proposition 1. ∎

The distortions in this paper therefore stem from pay-per-use pricing. A lump sum is unattractive when the firm cannot judge quality before using the service, when procurement is per defect, or when prepayment is costly.

Table 2. The incentive rent (H = 100, v = 1, k = 0.2).

Heterogeneity (a, b)	N^FB	W(N^FB)	G_min(N^FB)	Vendor's maximum at N^FB, R(N^FB)	Rent / W(N^FB)	Vendor's optimal depth ñ, R(ñ)
RepairBench-like (0.12, 0.39)	10	33.9	26.5	7.4	78%	3, 25.6
SWE-Verified-like (0.25, 0.34)	21	63.5	52.2	11.3	82%	4, 50.2
Synthetic (0.40, 3.0)	30	43.9	35.1	8.8	80%	10, 30.4

A vendor that wanted the efficient depth would leave the firm about four-fifths of the surplus.

4.4 Market structure

Private vendor quality. Suppose the vendor knows its type θ ∈ {H, L} with success parameters a_H > a_L (common b, k) and the firm has prior μ on H.

Proposition 7 (Profit-order reversal). For any common depth N ≥ 1, (a) under A the low type earns weakly more: (w−k)E_L(N) ≥ (w−k)E_H(N); (b) under B the high type earns strictly more when f > 0: fP_H(N) − kE_H(N) > fP_L(N) − kE_L(N).

Proof. The Beta family is ordered by likelihood ratio in *a*, so S(j) = E[(1−p)^j] is decreasing in a: S_H ≤ S_L, hence E_H ≤ E_L and P_H ≥ P_L, strictly for N ≥ 1 in P. ∎

Per-attempt pricing pays for failures, so it rewards the worse vendor at any depth, and a per-attempt contract cannot credibly signal quality; a success fee can. The intuition is that a less capable vendor needs

more attempts to deliver the same fix: when it is paid per attempt it bills those attempts, and when it is paid per fix it absorbs their cost. Numerically (Section 5.2), a high-quality vendor usually separates by *lowering* its fee below the full-information monopoly fee, which makes the firm sample deeper and partly corrects the monopoly under-sampling.

Competition among differentiated vendors. A unit mass of firms chooses among *n* symmetric vendors (same *a*, *b*, *k*) and human-only repair; firm *i*'s value from vendor *j* is G_j + σε_ij with Gumbel ε, where G_j is its saving at its optimal budget under j's contract and σ measures differentiation. Shares are logit, and vendors choose the price of their contract form in a symmetric Bertrand–Nash equilibrium.

Proposition 8 (Limits of competition). (a) As σ → 0 (close substitutes), per-attempt pricing converges to w = k and the efficient depth N^FB; success fees converge to the zero-profit fee f_0 ≤ f^e, so the firm over-samples (N_B(f_0) ≥ N^FB) and surplus is weakly below first-best. (b) Let ρ = $e^{\{S}$FB/σ} − 1. For σ large, (n+1)s_j ∈ [1−ρ, 1+ρ] at every price with positive profit, so every best response earns at least [(1−ρ)/(1+ρ)] of the monopoly profit per served firm; if the monopoly step is unique with margin Δ, the equilibrium budget equals the monopoly budget once σ > S^FB / ln(1 + Δ/(2Π_m − Δ)). Coverage tends to n/(n+1).

Proof. (a) With w = k the firm's threshold is the efficient one. For B, zero profit f_0 P = kE and participation vE + f_0P ≤ HP give (k+v)E ≤ HP and hence f_0 = kE/P ≤ Hk/(k+v). (b) With G ∈ [0, S^FB] at best responses, the share is (1+δ_j)/(n+1+Σδ_k) with δ ∈ [0, ρ]; the bounds follow, and a price outside the monopoly step earns at most (1+ρ)(Π_m − Δ)/(n+1) < (1−ρ)Π_m/(n+1) when ρ < Δ/(2Π_m − Δ). ∎

Differentiation, more than the number of vendors, determines whether competition removes the distortion; intermediate cases are solved numerically in Section 5.3.

Intuition. With logit demand, a vendor's share responds to its price in proportion to how much the price changes the firm's saving relative to the dispersion σ of firms' idiosyncratic preferences. When σ is small, firms barely prefer one vendor to another, a small price cut attracts everyone, and competition drives prices toward cost. When σ is large relative to the surplus, a price cut moves few firms, so each vendor

acts as a monopolist over its own customers and entry mostly divides the market: in Section 5.3, moving from one to ten vendors with σ = 30 raises coverage from 50% to 91% of firms, but the per-attempt budget stays at one attempt.

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5. Numerical Equilibria

Code in the replication package (anonymized for review). The shape parameters (a, b) come from the empirical fits in Section 6. Tables 2–4 use H = 100, v = 1, k = 0.2 (dollars), which lie inside the ranges calibrated from public sources in Section 5.4 (H) or between its function-level and agentic ranges (k); Section 5.4 shows how the conclusions change across those ranges. Integer budgets rule out closed forms for the vendor's optimal price, so equilibria are computed on price grids or by exact enumeration of budget steps.

5.1 Monopoly: per-attempt versus success fee

Table 3. Monopoly equilibria (H = 100, v = 1, k = 0.2). Entries: optimal price, firm's budget N, vendor profit, firm's saving, and welfare as a share of first-best.

Heterogeneity (a, b)	Per-attempt	Success fee
RepairBench-like (0.12, 0.39)	w* = 22.5, N=1, 22.3, 0.01, 66%	f* = 79.0, N=3, 25.6, 4.4, 89%
SWE-Verified-like (0.25, 0.34)	w* = 41.3, N=1, 41.1, 0.03, 65%	f* = 85.6, N=4, 50.2, 6.0, 88%
Synthetic (0.40, 3.0)	w* = 4.4, N=5, 17.3, 11.9, 66%	f* = 69.0, N=10, 30.4, 7.2, 85%

A per-attempt monopolist prices the *first* attempt ($w \approx q_0 H - v$) and the firm stops after one; the success fee extracts less per defect but sustains deeper sampling.

Over 296 random draws ($a \in [0.05, 0.6]$, $b \in [0.1, 4]$, $H \in [30, 500]$, $v/H \in [0.001, 0.03]$, $k/v \in [0.1, 3]$): the success fee raises vendor profit in 99% of draws and total surplus in 100% (median 91% of first-best, interquartile range 88–94%, versus 65%, 55–77%, under per-attempt pricing); median budgets are 28 (first-best), 1 (per-attempt), 6 (success fee). The firm's own saving is higher under the success fee in only 67% of draws, so outcome-based pricing can transfer surplus to the vendor and a side payment or two-part tariff may be needed.

5.2 Private quality

For a_H = 0.25, a_L = 0.08, b = 0.35, H = 100, v = 1, k = 0.2, the profit-order reversal is stark: at fee levels w = 10 and f = 50 and depth N = 10, the low type earns 73.8 under per-attempt pricing against the high type's 46.5, and under the success fee 14.8 against 32.6. In 60 random draws, the low type would mimic the high type's full-information contract in 98%; a separating linear contract exists in 77% (46 draws), always a success fee; separation costs the high type a median 11.0% of profit (interquartile range 7.1–16.8%); the induced budget is at least the full-information one in 76% of those draws, and total surplus and the firm's saving are higher under separation in 74% (median +7% and +9.7). In the remaining quarter the high type cannot separate by any linear contract and the market pools. Signaling by *generosity* thus partly offsets the monopoly distortion. These are numerical results under off-path beliefs that assign any other contract to the low type.

5.3 Competition

Table 4. Differentiated-Bertrand equilibria, RepairBench-like case (first-best N = 10). Entries: budget N, welfare as a share of first-best welfare for the same n and σ.

σ	Vendors n	Per-attempt: N, welfare	Success fee: N, welfare
1	2	9, 1.00	11, 1.00
10	2	3, 0.90	7, 0.97
10	5	4, 0.95	9, 1.00
30	2	1, 0.81	4, 0.94
30	5	1, 0.86	4, 0.96
30	10	1, 0.89	5, 0.98

With close substitutes (σ = 1) one rival suffices; with strongly differentiated vendors (σ = 30) the per-attempt equilibrium stays at the monopoly budget of one attempt even with ten vendors, and entry only widens coverage (from 50% to 91% of firms), as Proposition 8(b) predicts. The bound of Proposition 8(b) is conservative: the monopoly budget is reached at σ = 100. In 240 cells of a random sweep, the per-attempt equilibrium is not deviation-proof (a vendor gains by switching to a success fee) in 71% and the success-fee equilibrium is deviation-proof in 76%; vendors prefer the success fee in 55–100% of cells,

while the firm's saving is higher under per-attempt pricing in 68–98%. Vendors thus pick the contract form that raises surplus but that firms often like less.

5.4 Calibration of the cost parameters and sensitivity

Sources (accessed 25 October 2026; code the replication package (anonymized for review), `sensitivity.py`).

Human repair cost H. The US Bureau of Labor Statistics (Occupational Outlook Handbook) reports a median hourly wage of $64.44 for software developers in May 2025. Employer Costs for Employee Compensation (private industry, June 2022, the latest release we retrieved) put benefits at 29.5% of total compensation, so a loaded cost is $64.44 / (1 − 0.295) ≈ $91 per hour. The annotated time to fix for SWE-bench Verified (Epoch AI's summary of the annotation) is under 15 minutes for 39% of tasks, 15 minutes to 1 hour for 52%, 1–4 hours for 8%, and above 4 hours for 3 tasks; with bucket midpoints the expected hands-on time is 0.52 hours, hence H ≈ $48 for an engineer familiar with the code. METR's check found these estimates likely optimistic (its sample was only six tasks), so we allow 2–4 times that time for an unfamiliar engineer: **H ranges from about $15 (a 10-minute fix) through $50 to $200, up to $400 or more for multi-hour fixes**, before any context-transfer overhead.

Vendor cost per attempt k. We multiply observed token usage by the OpenRouter price list. For function-level repair in RepairBench, a non-reasoning attempt uses about 1,700 input and 800 output tokens, and a reasoning model such as o4-mini about 1,400 input and 10,400 output tokens. At list prices the cost k is $0.0001–0.0013 for small and mid-sized open models, $0.011 for a frontier model at 800 output tokens, and $0.0006–0.11 with reasoning-length outputs of 10,400 tokens borrowed from the o4-mini case ($0.047 for o4-mini itself). For repository-level agents, SWE-bench submission READMEs report $0.60 (Bracket), $0.64 (Moatless with Claude 4 Sonnet) and $0.84 (Cortexa) per task. **k ranges from $0.0003 to about $0.1 at function level and about $0.6–3 for agents.** (Our own HumanEvalFix runs billed $0.00002–0.00004 per attempt for 3–8B models.)

Firm cost per attempt v. If verification is an automated test run, v is the CI cost, about \$0.01–0.50. If an engineer spends 2–10 minutes triaging each candidate patch, v is about \$3–15 at \$91 per hour. This is the least well-grounded input; it is a judgement based on the loaded wage and not a measured quantity.

Sensitivity. We evaluate all combinations on a grid within each regime and three heterogeneity profiles (the RepairBench-like fit, the SWE-Verified-like calibration, and a HumanEvalFix-Llama-like fit with a = 0.50, b = 0.58), computing the monopoly equilibria exactly (profit at the top of each budget step; Appendix E). Table 5 summarizes, across the three profiles, the points at which AI repair is worth using ($N^{FB} \geq 1$).

Table 5. Sensitivity of the main conclusions across calibrated ranges (ranges across the three heterogeneity profiles; grids: 100–180 points per profile).

Regime (H, v, k grids)	AI repair worth using	Welfare: success fee > per-attempt	Vendor profit: success fee ≥ per-attempt	Firm saving: success fee > per-attempt	Median welfare / first best: per-attempt; success fee	Vendor's depth below N^{FB}	Median rent share of $W(N^{FB})$
Function-level, automated verification (H \$15–400; v \$0.01–0.5; k \$0.0003–0.1)	100%	97–100%	100%	97–100%	0.48–0.54; 0.87–0.93	100%	90–91%
Function-level, manual triage (H \$15–400; v \$3–15; k \$0.0003–0.1)	75–88%	22–57%	100%	22–57%	0.72–0.91; 0.87–0.91	61–86%	67–95%
Repository-level agents (H \$50–800; v \$0.1–5; k \$0.6–3)	100%	76–95%	100%	76–95%	0.56–0.67; 0.92–0.93	92–100%	37–44%

Findings. (i) The vendor's preference for the success fee is robust: its profit is at least as high in 100% of the points in every regime. (ii) The welfare advantage of the success fee is strong where an attempt is

cheap relative to human repair (automated verification: 97–100% of points; agentic repair: 76–95%) and weak when manual triage makes each attempt expensive (22–57%), because the efficient depth is then small (median 2–6) and a per-attempt monopolist that stops the firm after one or two attempts loses little. (iii) The incentive rent is large where depth is valuable, about nine-tenths of the surplus under automated verification and 37–44% for agents, so vendors undersample in 61–100% of points. (iv) Figure 2 shows the pattern in the cost ratios: when H/v is below about 30 the efficient depth is one or two attempts and both contract forms are near-efficient; above it, per-attempt monopoly welfare falls to 0.4–0.6 of first best while the success fee stays near 0.85–0.9. The conclusions of Sections 4–5 therefore apply to the regimes with H/v of about 30 or more, which covers automated verification and agentic repair and excludes cheap fixes with expensive per-attempt human review.

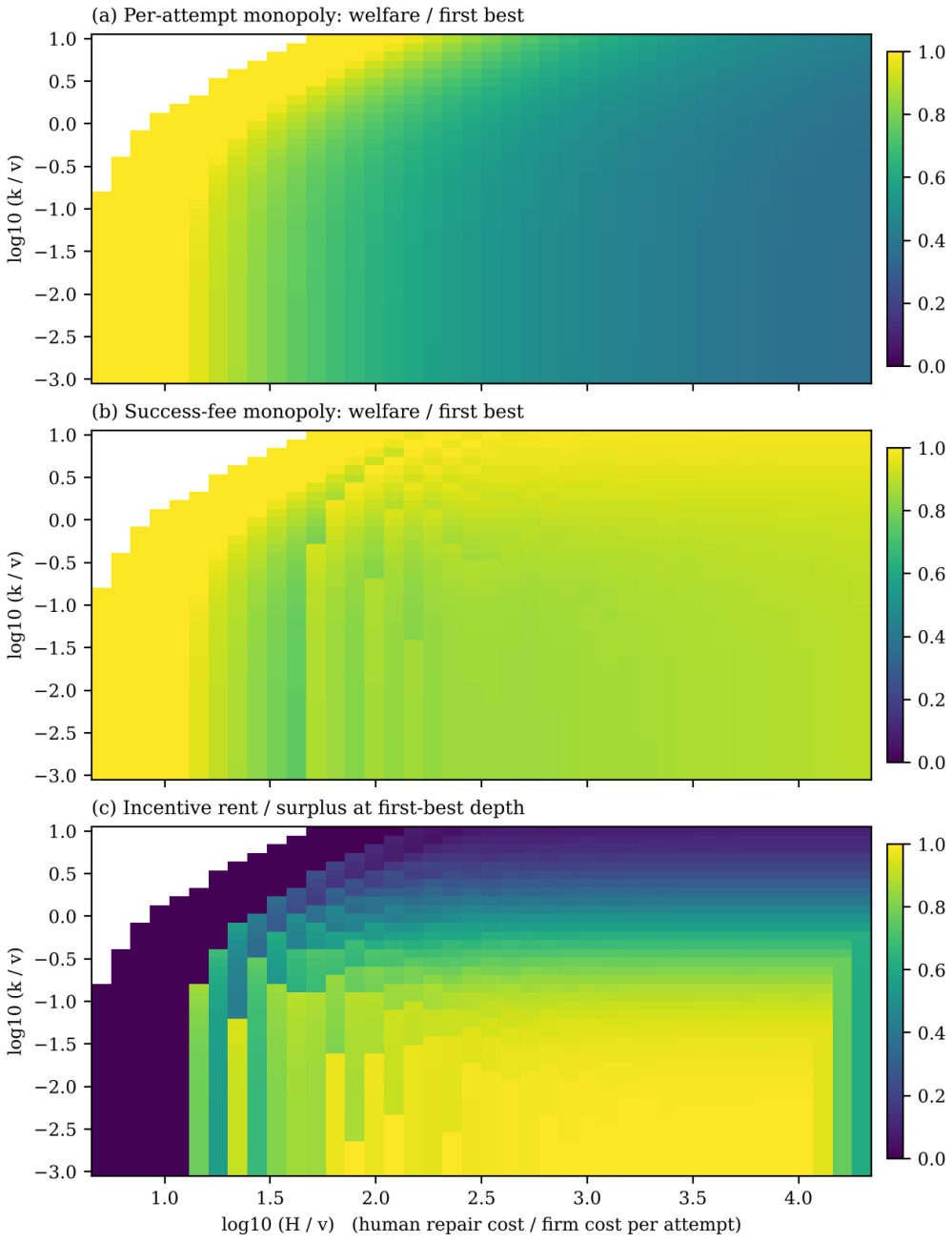


Figure 2. *Welfare (share of first best) under per-attempt and success-fee monopoly, and the incentive rent as a share of surplus at the first-best depth, as functions of H/v and k/v (a = 0.12, b = 0.39). Blank cells are combinations in which AI repair is not worth using.*

6. Evidence on the Model's Premises

The model rests on (P1) heterogeneous difficulty, (P2) informative failures, and (P3) good out-of-sample performance of a threshold fitted to history. Code in the replication package (anonymized for review).

RepairBench (Silva and Monperrus 2025): 574 single-function Java bugs (484 Defects4J v2, 90 GitBug-Java), ten sampled patches per bug and model, scored by the full test suite; 84 model × dataset combinations after requiring ten samples and at least 80 bugs (22,856 records). The Beta-binomial beats the homogeneous binomial in all 84 by AIC and held-out likelihood (50/50 split by bug identifier); median fitted $a = 0.12$, $b = 0.39$ (U-shaped), median 59% of bugs never solved in ten samples. Median thresholds at $H/c = 10, 30, 100, 300$ are 1, 3, 11, 34. Fitted on half the bugs and evaluated exactly on the other half (caps up to ten, cost in units of c), the reservation rule is within 0.4–1.7% of the best cap in hindsight and 34–67% below a ten-sample cap for H/c between 3 and 10; a ten-sample cap costs more than human-only repair when $H/c \leq 10$.

HumanEvalFix, 40 samples per bug (new; Python, 164 bugs; Llama-3.2-3B, Ministral-8B, Gemma-3-4B at temperature 1.0; 19,680 samples, $0.56 total). The Beta-binomial improves AIC over the binomial by thousands of points in all three (for example 7,293 to 739). A fit to the first ten samples predicts coverage at 40 samples within 2.6 points (for example Llama observed 0.927, predicted 0.909). For Gemma and Llama the fitted rule is within about 1% of the best cap. For Ministral, which solves 96% of bugs and leaves a small unsolvable residue, the rule oversamples ($N = 9$–40 against a hindsight best of 3–7) and costs 19–40% more than the best cap at $H/c \leq 30$, because the Beta tail assigns positive success probability to bugs that are never solved and half a sample of 164 bugs cannot identify that share. A firm should therefore validate the fitted threshold against a held-out cap sweep.

SWE-bench. The official submission logs are pass@1 by rule, so attempt-level data do not exist there. Two published pass@1/pass@k pairs identify the Beta parameters (calibration, not a test): DeepSWE (Verified) gives (0.25, 0.34) and the Lite series of Brown et al. (2024) gives (0.12, 0.65), again with $a < 1$. Coverage exceeds the verifier-selected rate (71.0% vs. 59.2%; 75% vs. 68.4%), so at repository level

success is observed only through an imperfect verifier, which lowers the benefit of success in the thresholds.

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7. Extensions

Imperfect verification. If a passing patch is wrong with probability e at downstream cost L, the benefit H in the thresholds becomes H − eL; weaker tests shorten the justified search, and in contract B disputes over what counts as a verified fix become a cost.

Further directions. Latency and severity-contingent thresholds, defect classes with their own (a, b), continuous vendor types and contract menus, multi-sourcing, dynamic learning of (a, b) across defects, and behavioral frictions on the buyer's side (for example loss aversion toward per-attempt payments or sunk-cost effects), which require experimental evidence to parameterize and are not modeled here.

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8. Discussion

Theory. (1) The economically relevant decision is whether to continue after a failure, not the pass@k capability. (2) A seller's markup distorts per-attempt contracts, and competition removes it only if vendors are close substitutes. (3) The underlying force is an incentive rent of sequential search, which explains why vendors prefer pure success fees, why they sample too little, and why a lump sum would restore efficiency. (4) Outcome-based pricing raises surplus, but not necessarily the buyer's share.

A buyer's procedure. (1) Sample each of a few hundred historical defects several times (ten samples per defect suffice) and record pass or fail. (2) Fit the Beta-binomial distribution of the counts to obtain (a, b). (3) Compute the cost per attempt c as the vendor's price plus the firm's own verification cost, and the threshold $N^* = \lceil aH/c - a - b \rceil$; with the running example (a = 0.12, b = 0.39, H = \$100), c = \$1.2 gives 10 attempts and c = \$6 gives 2. (4) Check the threshold against a sweep of caps on held-out defects, as the HumanEvalFix experiment shows that a fit can oversample when the share of unsolvable defects is small (Section 6). (5) Recompute when the vendor's price or the cost of human repair changes. If a vendor

quotes a price per fix, the same rule applies with the net value H − f in place of H and v as the cost of an attempt.

Managerial implications. *Buyers* should estimate (a, b) from their own repair logs, compute the threshold from H and the full cost per attempt (including verification), validate it on held-out defects, and negotiate a success fee near Hk/(k+v) in return for volume. *Vendors* gain from outcome-based pricing, including under competition, but should recognize that in commodity markets with close substitutes per-attempt pricing near cost is efficient. *Platforms and governance bodies* should publish per-sample outcomes and per-attempt costs, which determine the contracts' inputs and are invisible in current leaderboards.

Limitations. (1) H, v, and k are calibrated from public statistics and price lists (Section 5.4) but not measured on a firm, and v is the least well grounded; benchmark bugs (single functions with complete tests) are easier to verify than industrial defects. (2) Attempts are conditionally independent given p; correlation across samples would make failure more informative. (3) Vendors are symmetric, with logit competition and a one-shot game; private information is analyzed for two types and linear contracts, and the separation and competition results are numerical. (4) Patch quality is binary: a success fee pays for passing the tests, so a vendor paid per verified fix may neglect quality dimensions the tests do not measure (the multitask problem of Holmström and Milgrom 1991), which we do not model. (5) Public benchmarks may overlap with training data.

What would change the results. *Correlated attempts* (samples from the same prompt and model) would make failures more informative and lower the thresholds. *Imperfect verification* replaces H by H − eL in the thresholds, where e is the share of accepted patches that are wrong and L their downstream cost, and so also lowers them. *Learning about (a, b)* by the firm and the vendor over time would let the firm refine its rule and the vendor its contract. *Multi-sourcing* would give the firm a second price to compare, which acts like competition. *Vendor improvement* (a model that raises a over time) is outside the one-shot model.

Field validation. The analytical results do not depend on field data, but the contract predictions (outcome-based pricing under market power, per-attempt pricing among close substitutes) and the cost parameters would be tested directly by a study with a partner firm (Appendix D).

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9. Conclusion

An AI repair service creates two linked decisions: how long the buyer lets the AI try, and how the vendor charges for trying. When defects differ in difficulty, the buyer's optimal rule is a threshold that rises with the cost of human repair and falls with the price and difficulty of attempts. Charging per attempt above cost makes the buyer give up too early; a pure success fee is the vendor's best pay-per-use contract, but because deep sampling requires leaving the buyer an incentive rent, even it samples too little. A lump sum would remove every distortion; competition does so only among close substitutes. Public repair data support the premises of heterogeneous difficulty and informative failures, and a threshold fitted to past defects usually performs close to the best cap, though it can oversample when the unsolvable share is small and poorly identified. The next step is to test the contract predictions on a firm's repair and billing records.

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Appendix A. Survival Function and Posterior Predictive (Beta)

Given p, P(T > j | p) = (1−p)^j; integrating over p ~ Beta(a, b) gives S(j) = B(a, b+j)/B(a, b). After n failures the posterior is Beta(a, b+n) with mean a/(a+b+n) = q_n.

Appendix B. Estimation

With counts (n samples, k successes) per defect the Beta-binomial likelihood is C(n,k)·B(k+a, n−k+b)/B(a,b); first-success data censored at K attempts contribute B(a+1, b+t−1)/B(a,b) (solved at t) or B(a, b+K)/B(a,b) (unsolved). Both are maximized over (log a, log b) by Nelder–Mead (the replication package (anonymized for review)).

Appendix C. Reproducibility

- Contracts and equilibria: `cd paper/contracts && python3 contracts.py && python3 robustness.py`; private quality: `asymmetric.py`, `asym_sweep.py`; competition: `diff_bertrand.py`, `diff_sweep.py`; proof checks: `verify_proofs.py`; distribution-free checks (Propositions 1, 5, 6): `general_theory.py`.
- RepairBench: `cd paper/empirical && python3 load_repairbench.py <repairbench>/results && python3 analyze.py`; SWE-bench: `python3 swebench_repo_level.py <experiments>/evaluation`; HumanEvalFix: `cd paper/empirical/sampling && python3 sample_humanevalfix.py <model> 40 && python3 analyze_hef.py`.
- Cost calibration and sensitivity: `cd paper/contracts && python3 calibration.py && python3 sensitivity.py && python3 fig_sensitivity.py`.

Appendix D. Field-Data Request and Design (Not Executed)

Variables per defect: identifier and type; timestamps of arrival, each AI attempt, verification, escalation, closure; per attempt, test outcome, token counts, vendor-billed compute cost, verifier effort; escalation outcome; human engineering hours and fully loaded cost; post-merge defects traced to AI patches; contract terms. **Design:** randomize the attempt cap within strata (type, severity) for at least a quarter;

estimate (a, b) by stratum from the first part, compare realized cost per resolved defect under the reservation rule and the firm's existing cap; if the vendor offers both contract forms, randomize contract by team. **Confidentiality:** attempt-level outcomes and costs suffice; code and defect text are not needed.

Appendix E. Proofs

Lemma A (steps). For contract B, $N_B(f) = n \geq 1$ iff $q_n(H-f) \leq v < q_{n-1}(H-f)$, i.e. $f \in [\varphi_n, \varphi_{n-1})$ with $\varphi_n = H - v/q_n$ (Beta: $\varphi_n = H - v(n+a+b)/a$, equal step width v/a); $N_B(f) = 0$ iff $f \geq \varphi_0$. At $f = \varphi_n$ the firm is indifferent about attempt $n+1$ and stops.

Lemma B (boundary comparison). With $L(n) = \varphi_n P(n) - kE(n)$ and $R(n) = \varphi_{n-1}P(n) - kE(n)$, $R(n) - L(n-1) = \varphi_{n-1}[P(n) - P(n-1)] - k[E(n) - E(n-1)] = S(n-1)[\varphi_{n-1}q_{n-1} - k] = S(n-1)[q_{n-1}H - (k+v)]$, using $P(n) - P(n-1) = q_{n-1}S(n-1)$, $E(n) - E(n-1) = S(n-1)$, and $q_{n-1}(H - \varphi_{n-1}) = v$. The sign is that of the efficiency of attempt n, equivalently of $\varphi_{n-1} - f^e$.

Proof of Proposition 4. Profit within step n is $fP(n) - kE(n)$ with slope $P(n) > 0$, so $L(n) < \Pi < R(n)$ in the interior, and $R(n) - L(n) = (\varphi_{n-1} - \varphi_n)P(n) > 0$. Let $n_0 = N^{FB} \geq 1$, so $f^e \in [\varphi_{n_0}, \varphi_{n_0-1})$. *(b)* For $n > n_0$, $\varphi_{n-1} \leq \varphi_{n_0} \leq f^e$, so by Lemma B $R(n) \leq L(n-1) < R(n-1)$: R is strictly decreasing from n_0 on, hence $\hat{n} \leq n_0$. *(a)* Take $f' \leq f^e$ with $n = N_B(f') \geq n_0$. If $n = n_0$, any $f \in (f^e, \varphi_{n_0-1})$ has $\Pi(f) > \Pi(f')$ by monotonicity within the step. If $n > n_0$, $\Pi(f') < R(n) < R(n_0)$ and the supremum of Π on (f^e, φ_{n_0-1}) equals $R(n_0)$. ∎

Proof of Proposition 5. (a) Each term of G_{min} is nonnegative since $q_j \geq q_{N-1}$; $N = 1$ gives $P(1)/q_0 - E(1) = 1 - 1 = 0$; from N to N+1 existing terms weakly rise ($q_N \leq q_{N-1}$) and the new term is zero. (b) The firm chooses depth N only if $q_{N-1}(H-f) > w+v$, so $f < \varphi_{N-1} - w/q_{N-1}$. Then $\Pi = (w-k)E + fP < (w-k)E + [\varphi_{N-1} - w/q_{N-1}]P = R(N) - w[P/q_{N-1} - E]$, and $R(N) = W(N) - G_{min}(N)$ follows from $\varphi_{N-1}P - kE = HP - (k+v)E - v[P/q_{N-1} - E]$. (c) At $(0, \varphi_{N-1})$ participation holds since $vE + \varphi_{N-1}P \leq HP \Leftrightarrow q_{N-1}E \leq P$, true because $P = \Sigma_{j<N} S(j)q_j$. (d) $G_{min}(N+1) - G_{min}(N) = vP(N)(1/q_N - 1/q_{N-1})$ from $P(N+1) = P(N) + q_N S(N)$, $E(N+1) = E(N) + S(N)$, and $W(N+1) - W(N) = S(N)[q_N H - (k+v)]$. ∎